Operating Systems Placement Notes

Deadlock

Definition, Coffman conditions: Mutual Exclusion, Hold and Wait, No Preemption, Circular Wait. Prevention breaks a condition. Avoidance keeps system in safe state using Banker's Algorithm. Detection allows deadlock then recovers.

Banker's Algorithm

Need = Max - Allocation. Grant a request only if Request<=Need, Request<=Available and the resulting state is safe.

Memory Management

Logical vs Physical Address, MMU, contiguous vs non-contiguous allocation.

Paging

Fixed-size pages and frames, page table, no external fragmentation, internal fragmentation possible.

Segmentation

Variable-sized logical segments (code, data, stack), segment table, external fragmentation possible.

Paging vs Segmentation

Paging=fixed hardware view; Segmentation=variable logical view.

Virtual Memory

Allows programs larger than RAM using disk. Demand paging loads pages on demand. Page fault occurs when page absent in RAM.

Swapping

Moves an entire process between RAM and disk. Paging moves individual pages.

Page Replacement

FIFO: oldest page replaced. LRU: least recently used page replaced. Optimal: replace page used farthest in future (theoretical). Clock/Second Chance approximates LRU in practice.

Interview Q&A;

Q: What is deadlock? A: Processes wait forever for each other's resources. Q: Four deadlock conditions? A: Mutual exclusion, Hold & Wait, No Preemption, Circular Wait. Q: Difference between prevention and avoidance? A: Prevention breaks a condition; avoidance checks safe state. Q: What is paging? A: Fixed-size pages mapped to frames. Q: Paging vs segmentation? A: Fixed vs variable, physical vs logical. Q: What is virtual memory? A: Disk extension of RAM. Q: What is a page fault? A: Requested page not in RAM. Q: FIFO vs LRU? A: FIFO removes oldest page; LRU removes least recently used page. Q: Best page replacement? A: Optimal theoretically; LRU/Clock-like methods in

practice. Q: Which algorithm is used in industry? A: LRU approximations such as Clock/Second Chance and advanced variants.